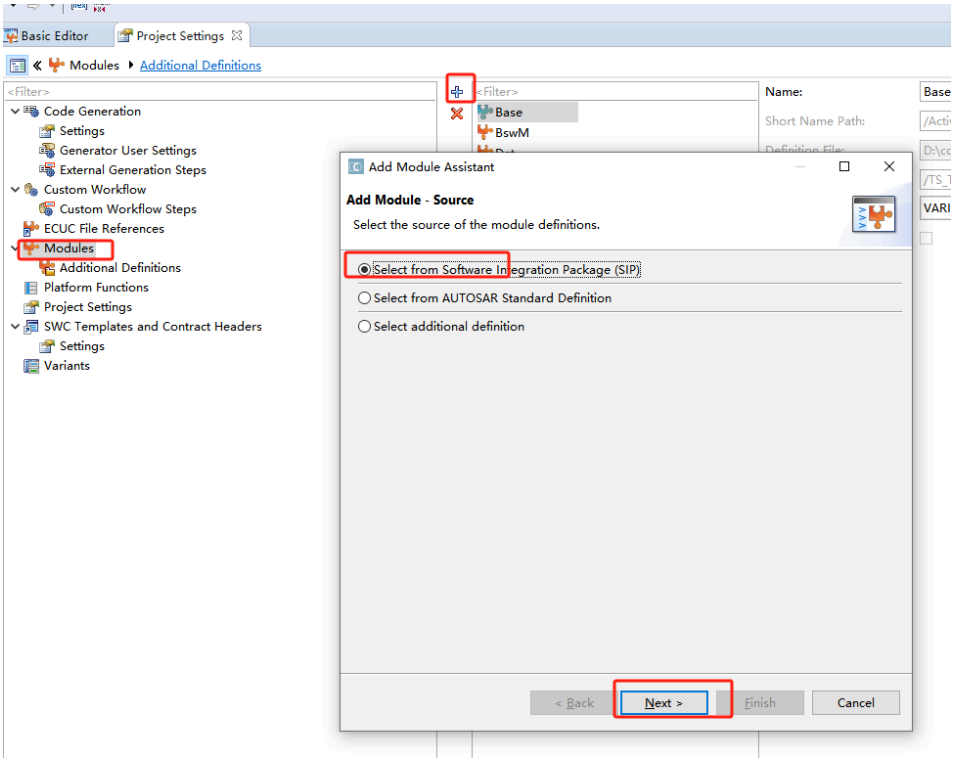
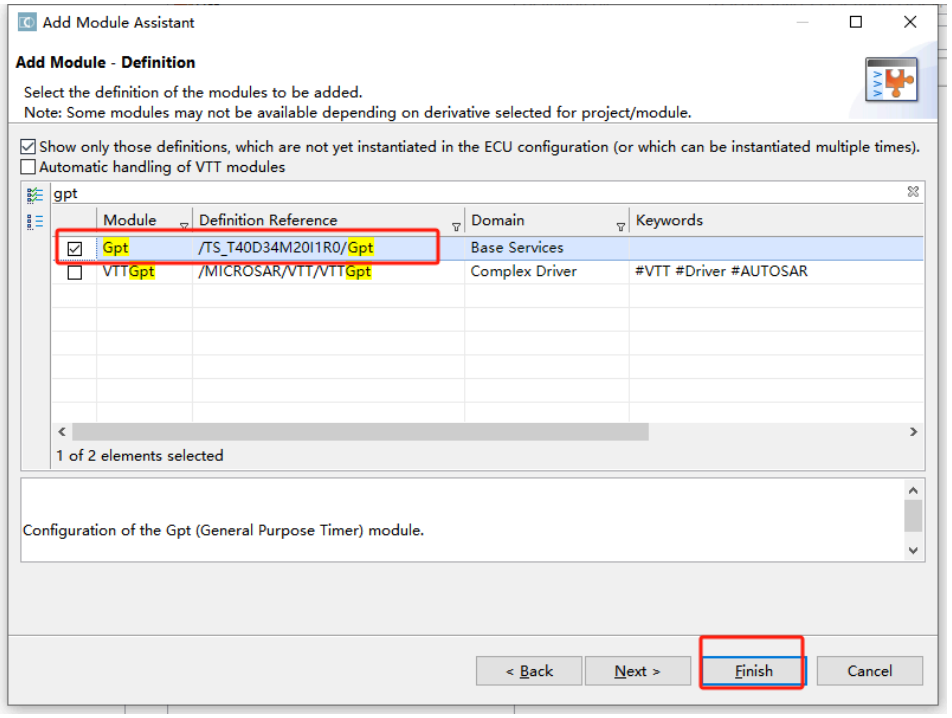
8、新增Gpt模块

卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

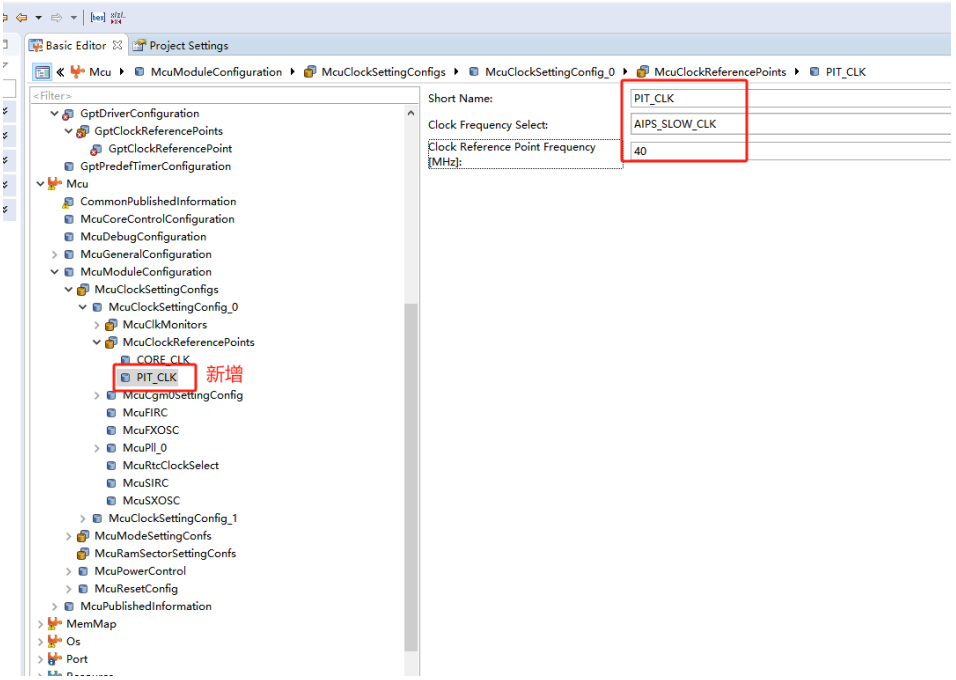
一、添加Gpt到davinci cfg



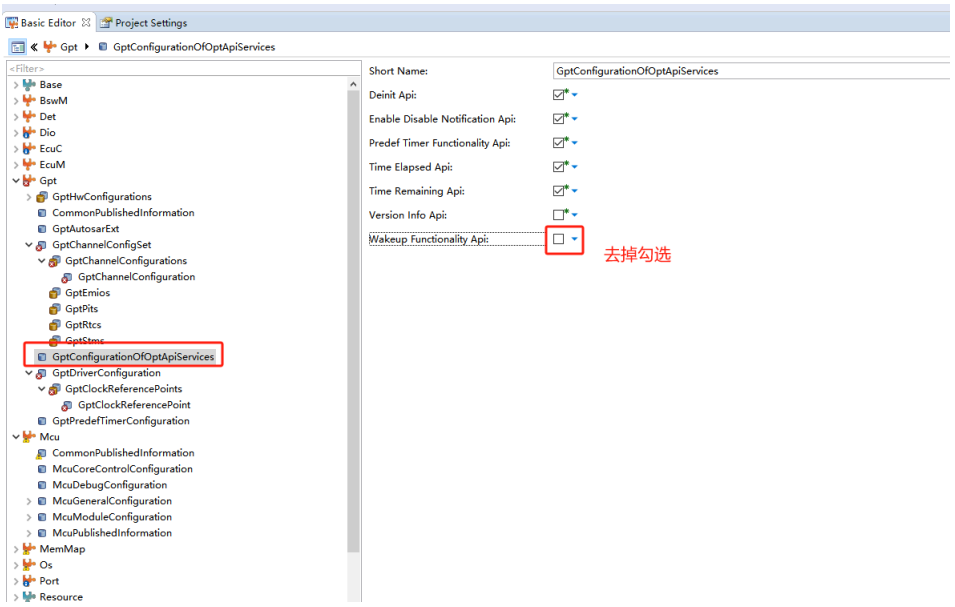


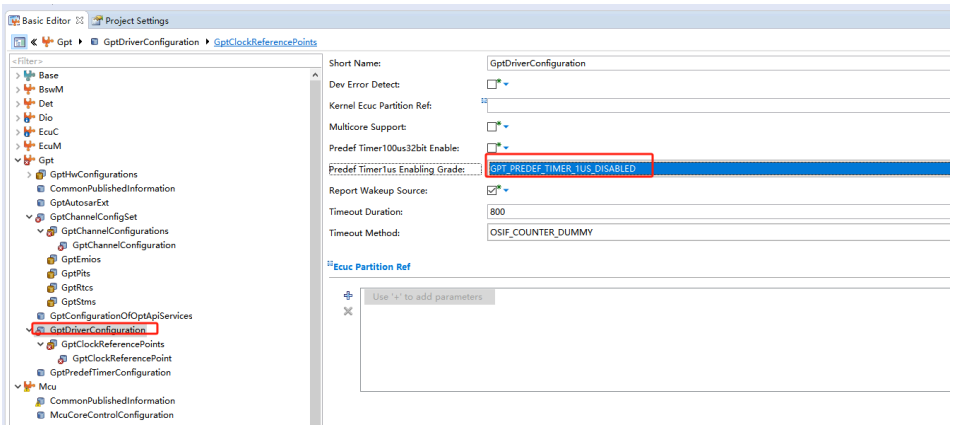
二、消除错误

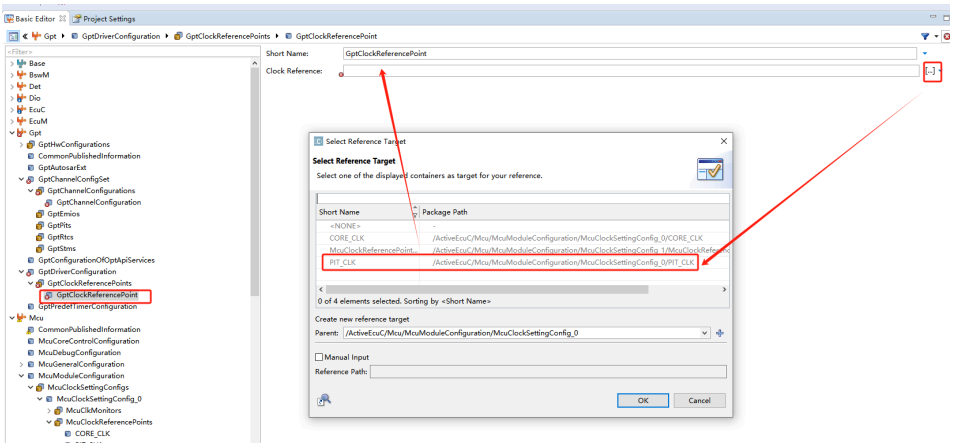
第一步先在MCU模块新增一PIT的时钟

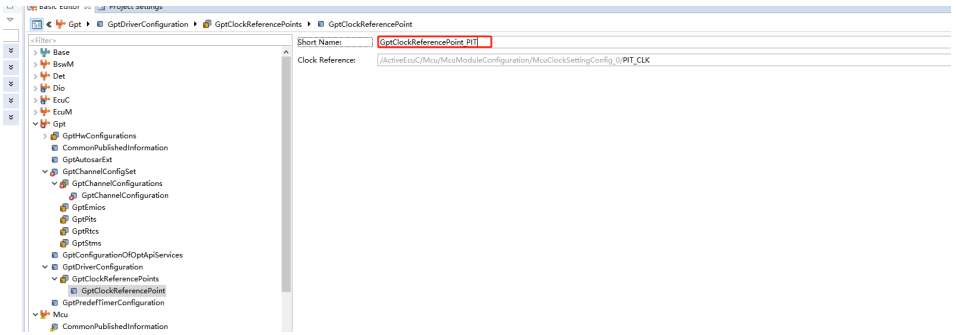


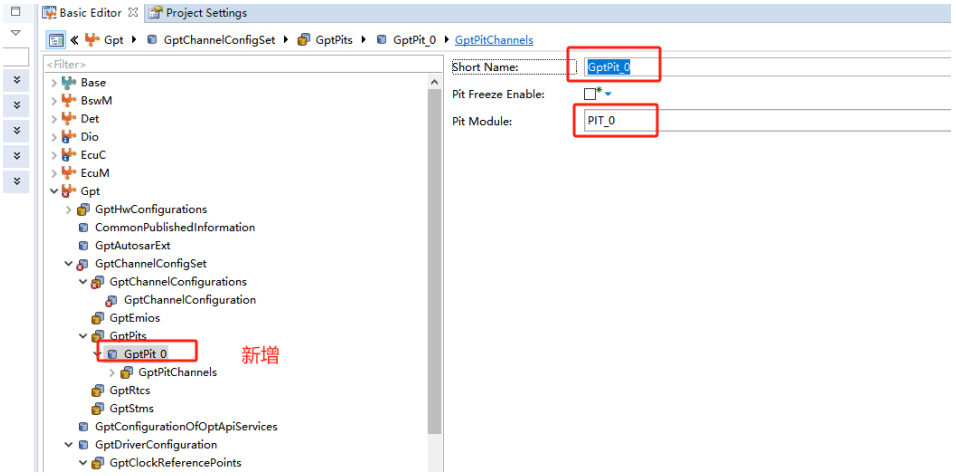
gpt消除错误

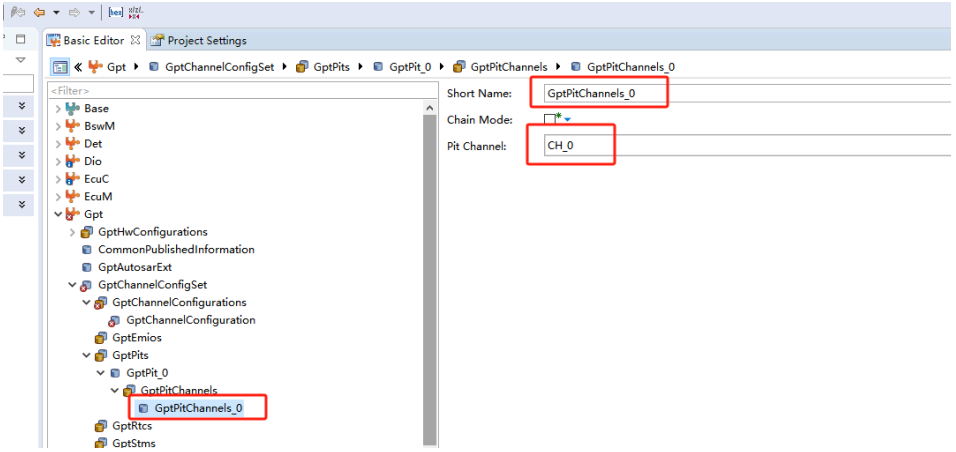

Basic Editor
Project Settings
Gpt ▸ GptConfigurationOfOptApiServices
Short Name:
GptConfigurationOfOptApiServices
Deinit Api:
Enable Disable Notification Api:
Predef Timer Functionality Api:
Time Elapsed Api:
Time Remaining Api:
Version Info Api:
Wakeup Functionality Api:
去掉勾选
GptConfigurationOfOptApiServices
GptDriverConfiguration
GptClockReferencePoints
GptClockReferencePoint
GptPredefTimerConfiguration


Basic Editor
Project Settings
Gpt ▸ GptDriverConfiguration ▸ GptClockReferencePoints
Short Name:
GptDriverConfiguration
Dev Error Detect:
Kernel Ecuc Partition Ref:
Multicore Support:
Predef Timer100us32bit Enable:
Predef Timer1us Enabling Grade:
GPT_PREDEF_TIMER_1US_DISABLED
Report Wakeup Source:
Timeout Duration:
800
Timeout Method:
OSIF_COUNTER_DUMMY
Ecuc Partition Ref
Use '+' to add parameters

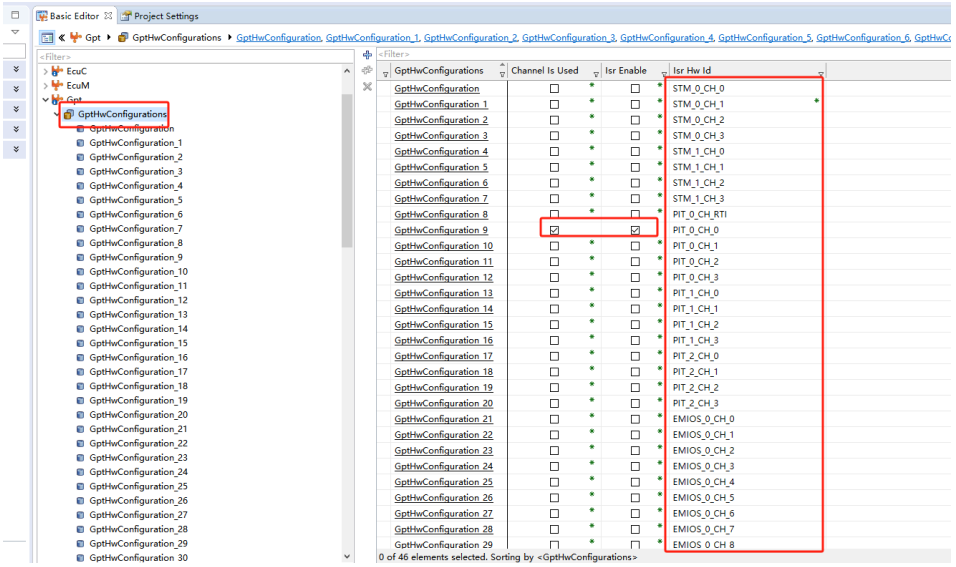

Basic Editor
Project Settings
Gpt ▸ GptDriverConfiguration ▸ GptClockReferencePoints ▸ GptClockReferencePoint
Short Name:
GptClockReferencePoint
Clock Reference:
Select Reference Target
Select Reference Target
Select one of the displayed containers as target for your reference.
Short Name
Package Path
CORE_CLK
PIT_CLK
Create new reference target
Parent:
Manual Input
Reference Path:
OK
Cancel

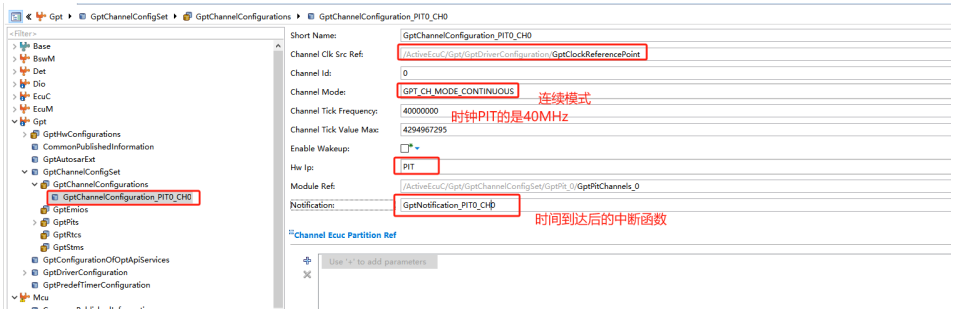






GptHwConfiguration 按照顺序将46个都选择以下，然后用到的PIT_0_CH_0 勾选used 和isr enable





三、OS 添加中断

中断函数名可以在mcal代码中找到

CtApTestSwc.c | BswM_Lcfg.c | EcuM_Callout_Stubs.c | ARMStartup_CortexM.c | **Pit_Ip.c** | Pit_Ip_Cfg_Defines.h | MemMap.h | vLinkGen_Template

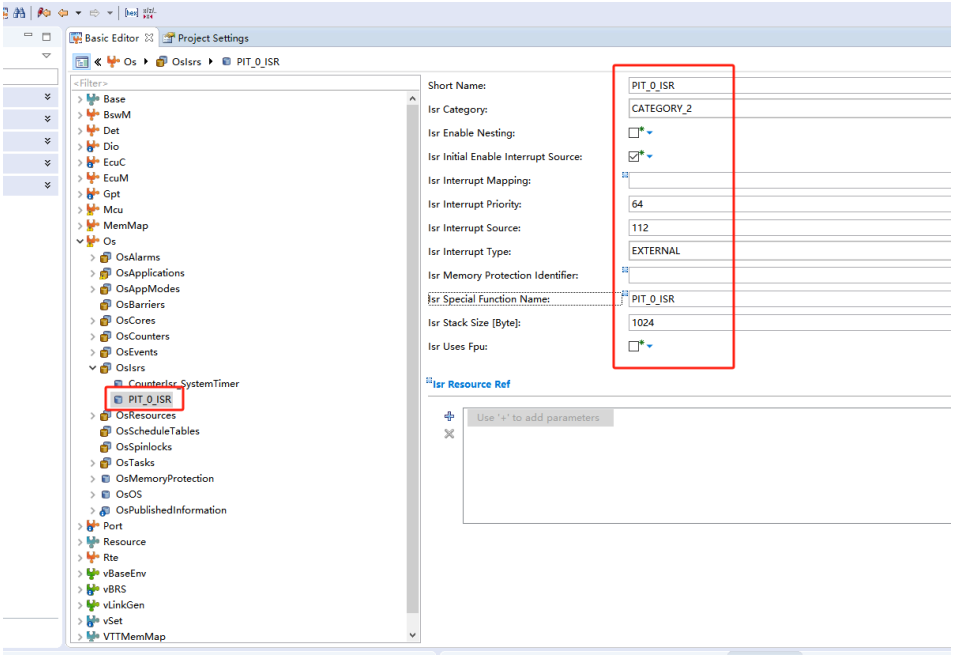
Os_Isr_PIT_0_ISR()

```
    return lifeTimeValue;
}
/*==================================================================================================*/
#if (PIT_IP_MODULE_SINGLE_INTERRUPT == STD_ON)
#ifdef PIT_0_ISR_USED
/**
* @brief   Interrupt handler for PIT_0 channels.
* @details Interrupt Service Routine corresponding to PIT_0 hw module.
* @param[in] none
* @return  void
* @isr
* @pre      The driver needs to be initialized
*/
ISR(PIT_0_ISR)
{
#if ((defined PIT_IP_PECULIAR_INSTANCES) && (PIT_IP_PECULIAR_INSTANCES == STD_ON))
    uint8 instance = 6U;
#else
    uint8 instance = 0U;
#endif
    uint8 channel = 0U;

#if ((defined PIT_IP_RTI_CHANNEL_EXISTS) && (PIT_IP_RTI_USED == STD_OFF))

    for (channel = 0U; channel < PIT_0_IP_CHANNELS_NUMBER - 1U; channel++)
#else
    for (channel = 0U; channel < (PIT_0_IP_CHANNELS_NUMBER); channel++)
#endif
    {
        Pit_Ip_ProcessCommonInterrupt(instance, channel);
    }
}
#endif
/*==================================================================================================*/
#ifdef PIT_1_ISR_USED
/**
* @brief   Interrupt handler for PIT_1 channels.
* @details Interrupt Service Routine corresponding to PIT_1 hw module.
```

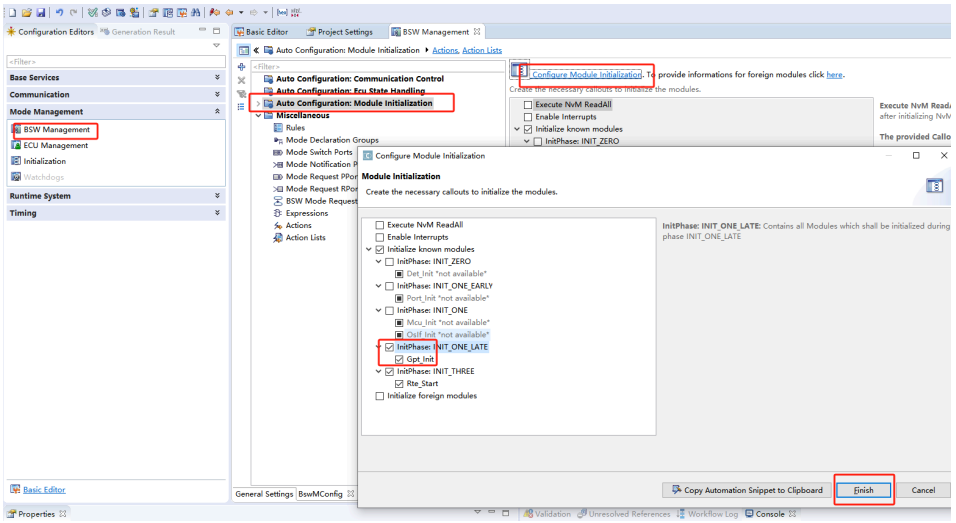
中断号

G189 | fx | PIT_0

	A	B	C	D	E	F	G	
1	Address	Vector (all cores)	IRQ	NVIC IPR register number	NVIC non-IPR register number	INTM support	Instance	Interrupt request
160								Interrupts from pad group 1
161								Interrupts from pad group 2
162								Interrupts from pad group 3
163								Interrupts from pad group 0_64
164								Interrupts from pad group 1_64
165								Interrupts from pad group 2_64
166								Interrupts from pad group 3_64
167	0x00000190	100	84	21	2	Y	CMU_0	CMU0 interrupt
168	0x00000194	101	85	21	2	Y	CMU_1	CMU1 interrupt
169	0x00000198	102	86	21	2	Y	CMU_2	CMU2 interrupt
170	0x0000019C	103	87	21	2	Y	Body Cross Triggering Unit	An interrupt is requested when a conversio
171								An interrupt is requested when new data is
172								An interrupt is requested when new data is
173								An interrupt is requested when new data is
174								An interrupt is requested when the last co
175								An Interrupt output for FIFO1
176								An Interrupt output for FIFO2
177	0x000001A0	104	88	22	2		RESERVED	
178	0x000001A4	105	89	22	2		RESERVED	
179	0x000001A8	106	90	22	2		RESERVED	
180	0x000001AC	107	91	22	2		RESERVED	
181	0x000001B0	108	92	23	2	Y	Logic Control Unit 0	Interrupt_0
182								Interrupt_1
183								Interrupt_2
184	0x000001B4	109	93	23	2	Y	Logic Control Unit 1	Interrupt_0
185								Interrupt_1
186								Interrupt_2
187	0x000001B8	110	94	23	2		RESERVED	
188	0x000001BC	111	95	23	2		RESERVED	
189	0x000001C0	112	96	24	3	Y	PIT_0	Interrupt for Channel0
190								Interrupt for Channel1
191								Interrupt for Channel2
192								Interrupt for Channel3
193								Interrupt for Channel4
194	0x000001C4	113	97	24	3	Y	PIT_1	Interrupt for Channel0
195								Interrupt for Channel1
196								Interrupt for Channel2
197								Interrupt for Channel3
198	0x000001C8	114	98	24	3	Y	PIT_2	Interrupt for Channel0
199								Interrupt for Channel1
200								Interrupt for Channel2
201								Interrupt for Channel3
202	0x000001CC	115	99	24	3		RESERVED	



四、BswM添加Gpt的init

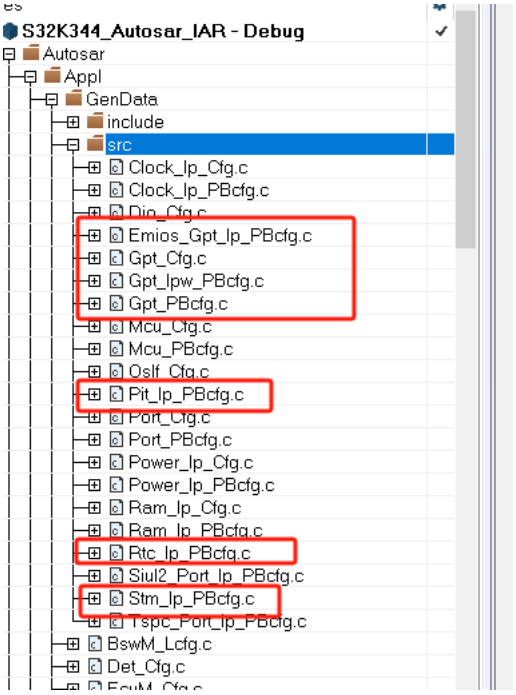


五、生成Gpt、Os、Mcu、Rte、vLinkGen

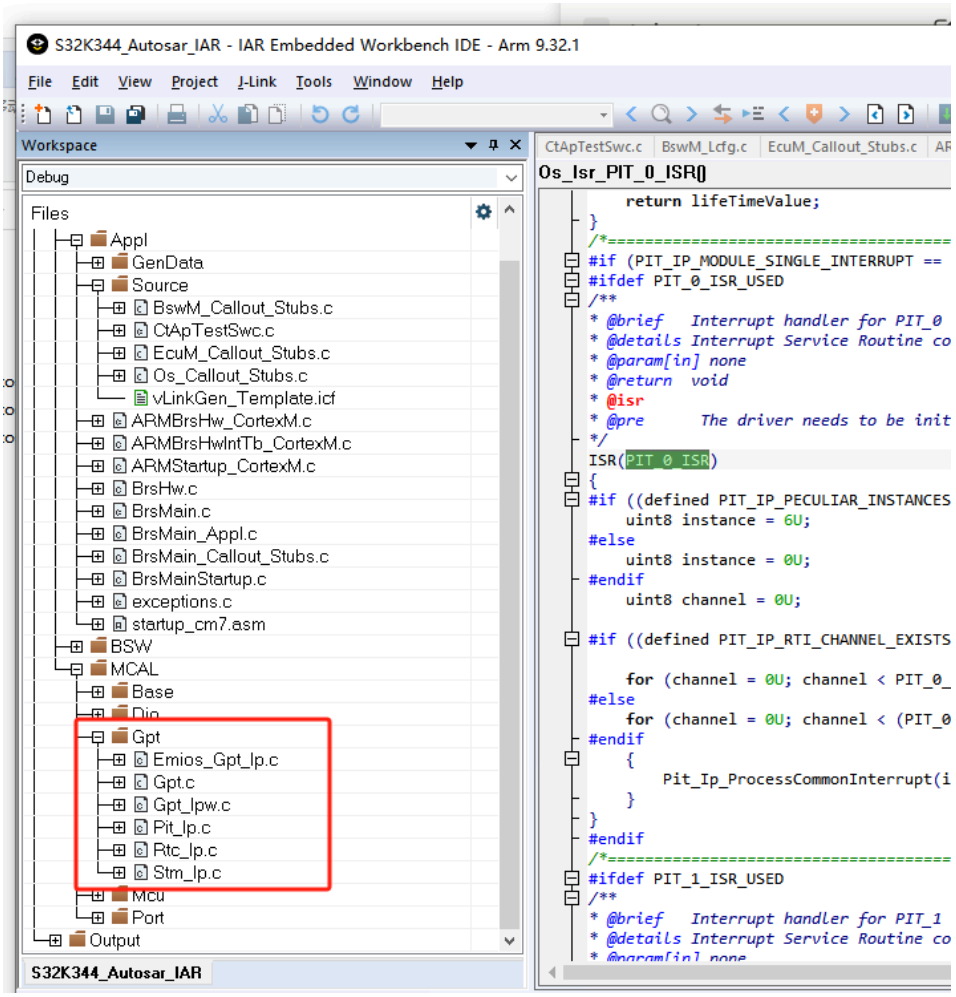
六、IAR添加MCAL gpt代码

先把CBD2300384_D00\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_2.0.1\eclipse\plugins\Base_TS_T40D34M20I1R0\inc下的_Gpt_MemMap.h 改名为Gpt_MemMap.h

添加davinci 生成的gpt代码

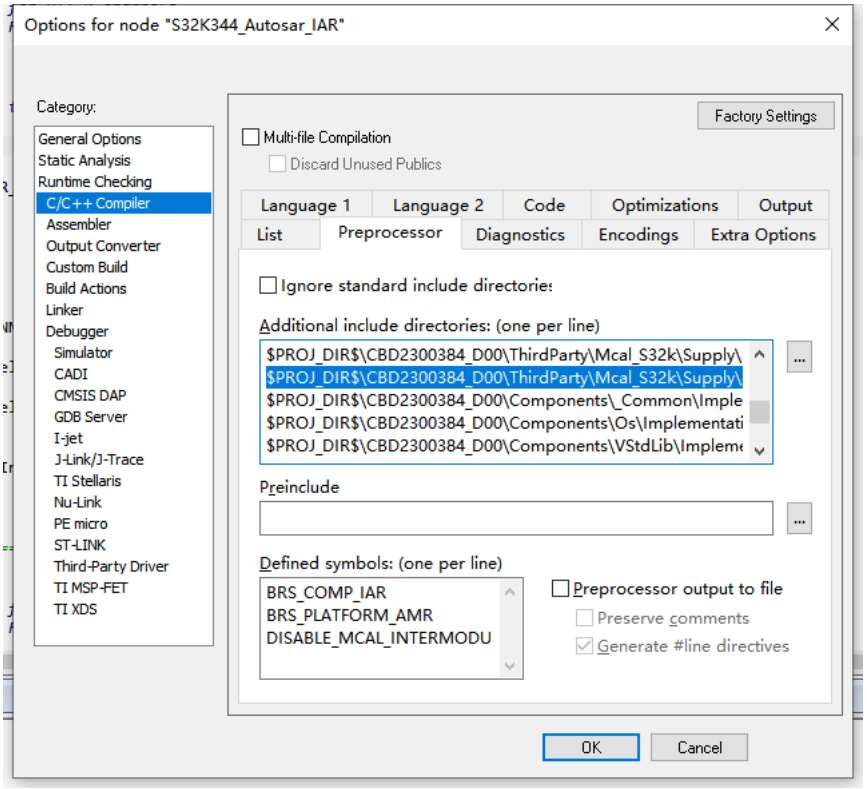

S32K344_Autosar_IAR - Debug
Autosar
Appl
GenData
include
src
Clock_Ip_Cfg.c
Clock_Ip_PBcfg.c
Dio_Cfg.c
Emios_Gpt_Ip_PBcfg.c
Gpt_Cfg.c
Gpt_Ipw_PBcfg.c
Gpt_PBcfg.c
Mcu_Cfg.c
Mcu_PBcfg.c
Oslf_Cfg.c
Pit_Ip_PBcfg.c
Port_Cfg.c
Port_PBcfg.c
Power_Ip_Cfg.c
Power_Ip_PBcfg.c
Ram_Ip_Cfg.c
Ram_Ip_PBcfg.c
Rtc_Ip_PBcfg.c
Siul2_Port_Ip_PBcfg.c
Stm_Ip_PBcfg.c
Tspc_Port_Ip_PBcfg.c
BswM_Lcfg.c
Det_Cfg.c

添加MCAL的gpt


S32K344_Autosar_IAR - IAR Embedded Workbench IDE - Arm 9.32.1
File Edit View Project J-Link Tools Window Help
Workspace
Debug
Files
Appl
GenData
Source
BswM_Callout_Stubs.c
CtApTestSwc.c
EcuM_Callout_Stubs.c
Os_Callout_Stubs.c
vLinkGen_Template.icf
ARMBrsHw_CortexM.c
ARMBrsHwIntTb_CortexM.c
ARMStartup_CortexM.c
BrsHw.c
BrsMain.c
BrsMain_Appl.c
BrsMain_Callout_Stubs.c
BrsMainStartup.c
exceptions.c
startup_cm7.asm
BSW
MCAL
Base
Dio
Gpt
Emios_Gpt_Ip.c
Gpt.c
Gpt_Ipw.c
Pit_Ip.c
Rtc_Ip.c
Stm_Ip.c
Mcu
Port
Output
S32K344_Autosar_IAR
CtApTestSwc.c
BswM_Lcfg.c
EcuM_Callout_Stubs.c
Os_Isr_PIT_0_ISR()
return lifeTimeValue;
}
#if (PIT_IP_MODULE_SINGLE_INTERRUPT ==
#ifdef PIT_0_ISR_USED
/**
* @brief Interrupt handler for PIT_0
* @details Interrupt Service Routine co
* @param[in] none
* @return void
* @isr
* @pre The driver needs to be init
*/
ISR(PIT_0_ISR)
{
#if ((defined PIT_IP_PECULIAR_INSTANCES
uint8 instance = 6U;
#else
uint8 instance = 0U;
#endif
uint8 channel = 0U;
#if ((defined PIT_IP_RTI_CHANNEL_EXISTS
for (channel = 0U; channel < PIT_0_
#else
for (channel = 0U; channel < (PIT_0
#endif
{
Pit_Ip_ProcessCommonInterrupt(i
}
}
#endif
#ifdef PIT_1_ISR_USED
/**
* @brief Interrupt handler for PIT_1
* @details Interrupt Service Routine co

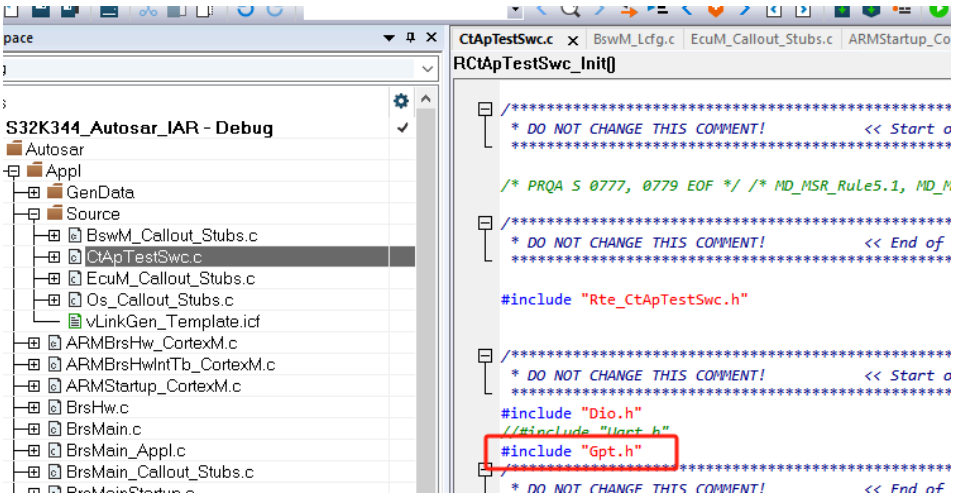
添加MCAL gpt的头文件路径在工程

$PROJ_DIR$\CBD2300384_D00\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_2.0.1\eclipse\plugins\Gpt_TS_T40D34

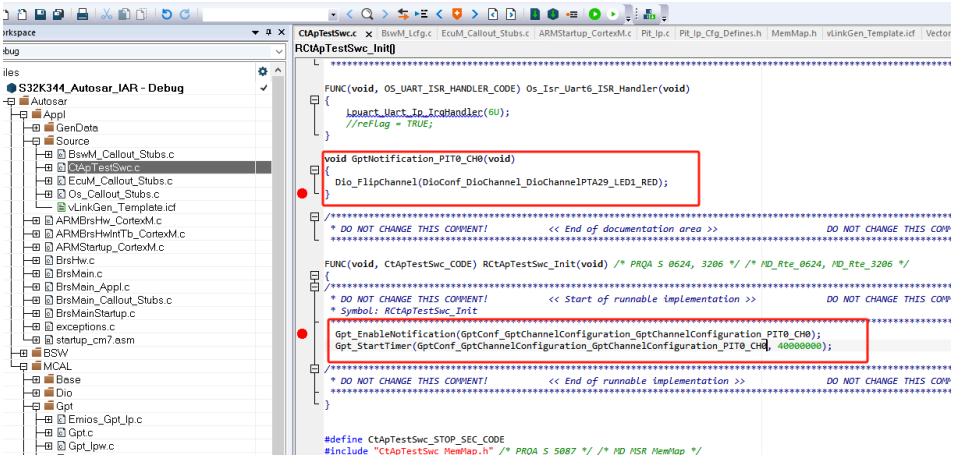


七、添加测试代码

头文件



在定时1s到了之后翻转一次IO




```
void GptNotification_PIT0_CH0(void)
{
Dio_FlipChannel(DioConf_DioChannel_DioChannelPTA29_LED1_RED);
}
Gpt_EnableNotification(GptConf_GptChannelConfiguration_GptChannelConfiguration_PIT0_CH0);
Gpt_StartTimer(GptConf_GptChannelConfiguration_GptChannelConfiguration_PIT0_CH0, 40000000);
```